

Notebook

March 22, 2025

```
[1]: import plotly.express as px
import pandas as pd
import plotly.figure_factory as ff
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
[2]: df = pd.read_csv('heart.csv')
```

```
[3]: df
```

```
[3]:
```

	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	\
0	63	1	3	145	233	1	0	150	0	2.3	0	
1	37	1	2	130	250	0	1	187	0	3.5	0	
2	41	0	1	130	204	0	0	172	0	1.4	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	
..	...	...	..	...	...	...	...	...	...	...	...	
298	57	0	0	140	241	0	1	123	1	0.2	1	
299	45	1	3	110	264	0	1	132	0	1.2	1	
300	68	1	0	144	193	1	1	141	0	3.4	1	
301	57	1	0	130	131	0	1	115	1	1.2	1	
302	57	0	1	130	236	0	0	174	0	0.0	1	

	caa	thall	output
0	0	1	1
1	0	2	1
2	0	2	1
3	0	2	1
4	0	2	1
..	...	...	...
298	0	3	0
299	0	3	0
300	2	3	0
301	1	3	0
302	1	2	0

[303 rows x 14 columns]

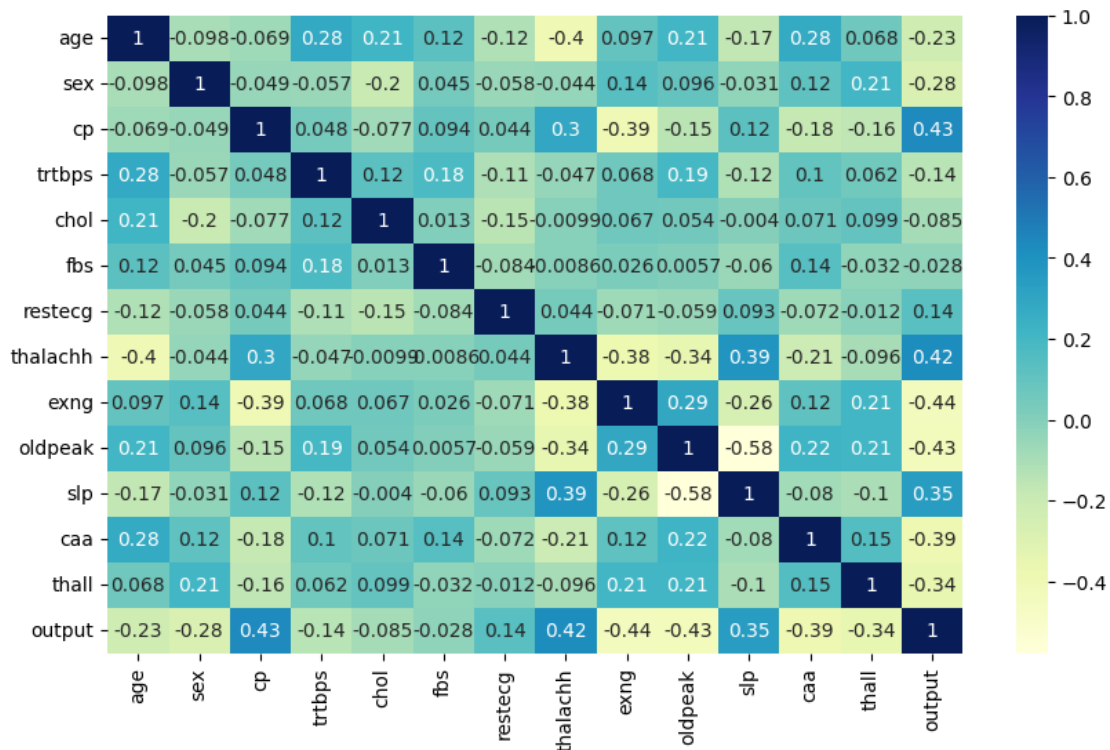
```
[4]: df.isna().sum()
```

```
[4]: age      0
     sex      0
     cp       0
     trtbps   0
     chol     0
     fbs      0
     restecg  0
     thalachh 0
     exng     0
     oldpeak  0
     slp      0
     caa      0
     thall    0
     output   0
     dtype: int64
```

0.1 1)Heatmap

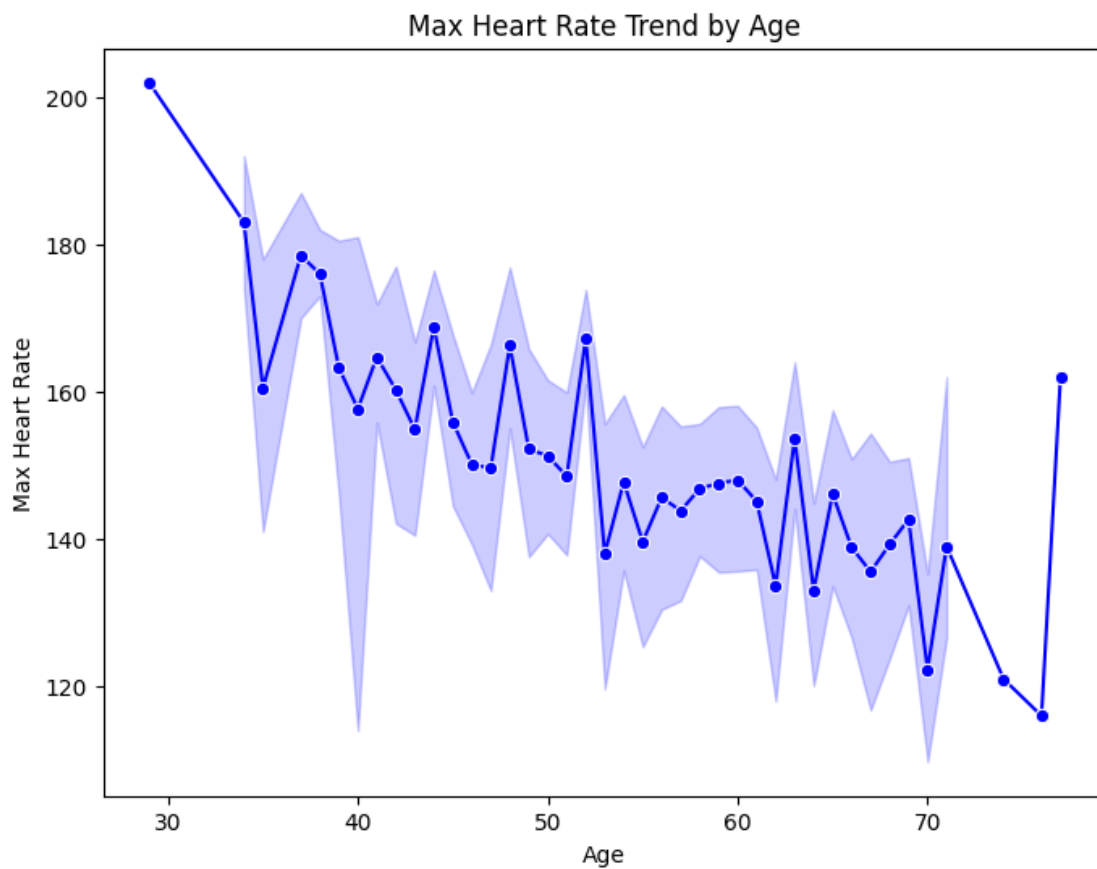
```
[5]: plt.figure(figsize=(10,6))
     sns.heatmap(df.corr(),cmap = 'YlGnBu', annot = True)
```

```
[5]: <Axes: >
```



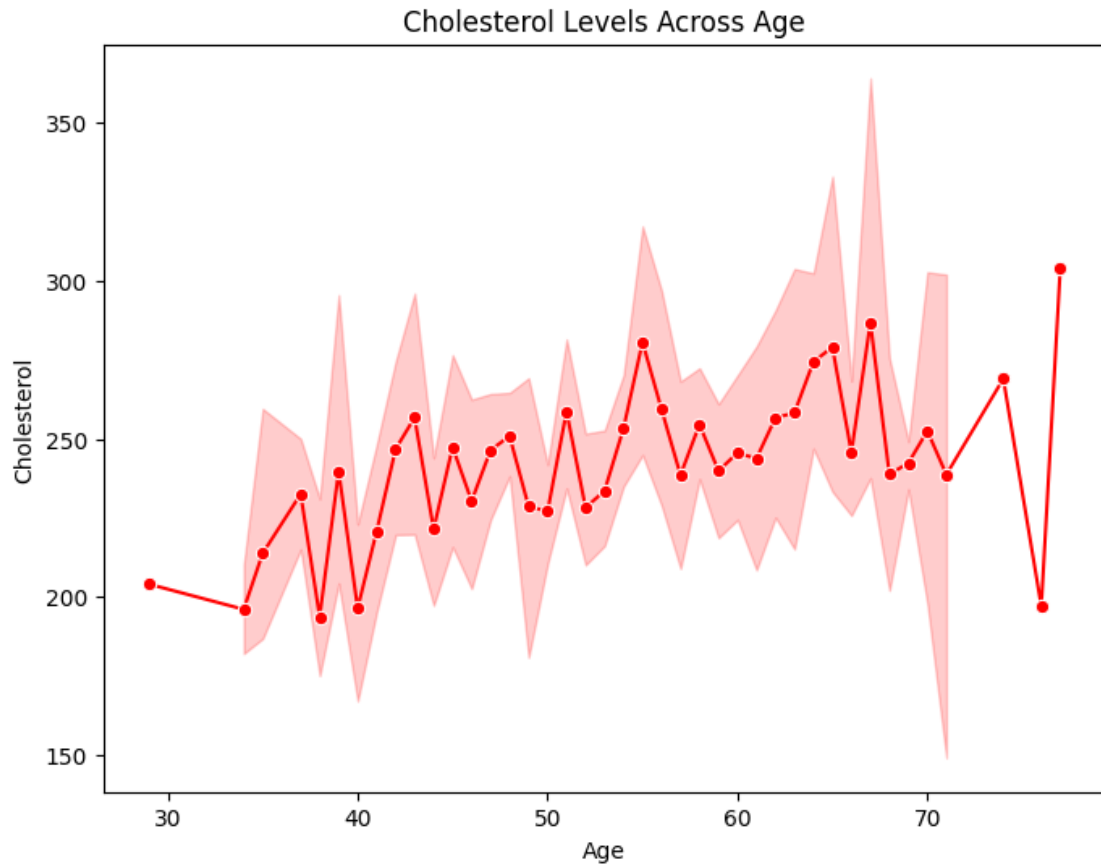
0.2 2)Line Plots

```
[6]: # Max Heart Rate Trend by Age
plt.figure(figsize=(8,6))
sns.lineplot(data=df.sort_values(by="age"),
             x="age", y="thalachh", marker="o",
             color="blue")
plt.title("Max Heart Rate Trend by Age")
plt.xlabel("Age")
plt.ylabel("Max Heart Rate")
plt.show()
```



```
[7]: # Cholesterol Levels Across Age
plt.figure(figsize=(8,6))
sns.lineplot(data=df.sort_values(by="age"),
             x="age", y="chol", marker="o",
             color="red")
```

```
plt.title("Cholesterol Levels Across Age")
plt.xlabel("Age")
plt.ylabel("Cholesterol")
plt.show()
```



```
[8]: # Resting Blood Pressure Trend by Age
plt.figure(figsize=(8,6))
sns.lineplot(data=df.sort_values(by="age"),
             x="age", y="trtbps", marker="o",
             color="green")
plt.title("Resting Blood Pressure Across Age")
plt.xlabel("Age")
plt.ylabel("Resting Blood Pressure")
plt.show()
```

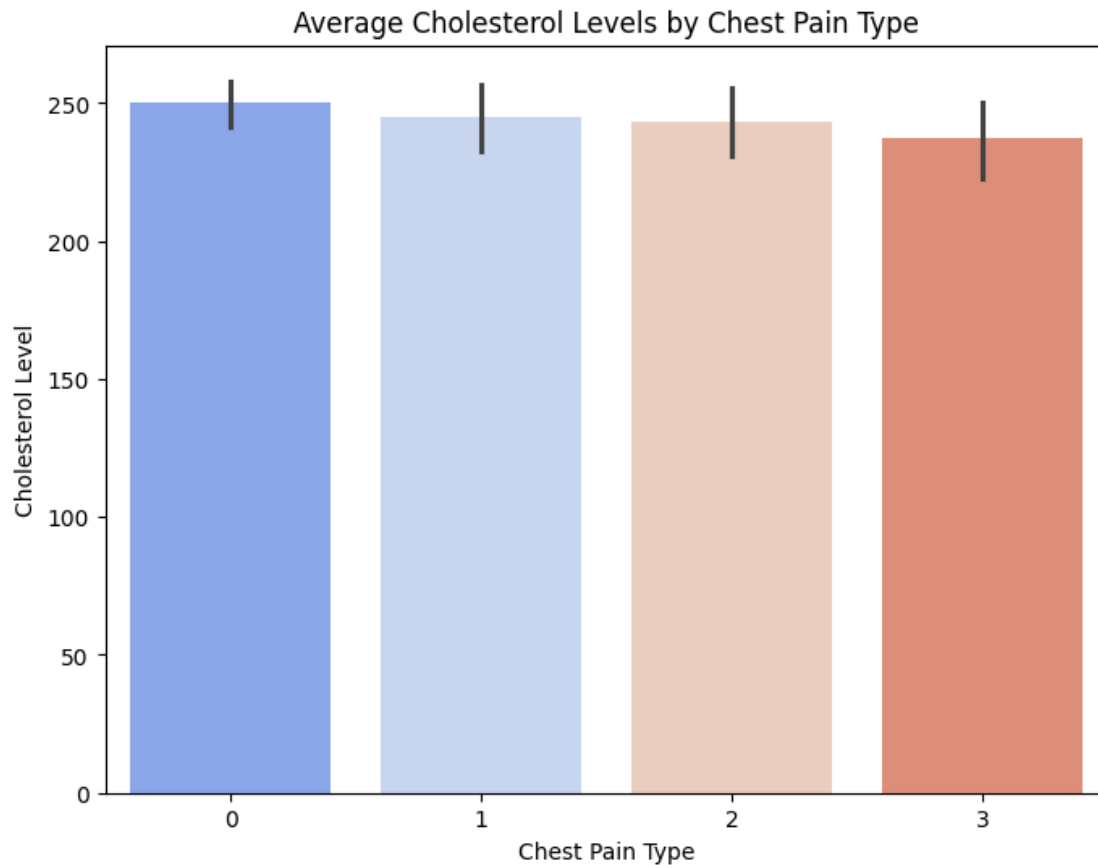


0.3 3)Bar plot

```
[9]: # Average Cholesterol Levels by Chest Pain Type
plt.figure(figsize=(8,6))
sns.barplot(data=df, x="cp", y="chol", palette="coolwarm")
plt.title("Average Cholesterol Levels by Chest Pain Type")
plt.xlabel("Chest Pain Type")
plt.ylabel("Cholesterol Level")
plt.show()
```

C:\Users\Yash\AppData\Local\Temp\ipykernel_54992\3822819362.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

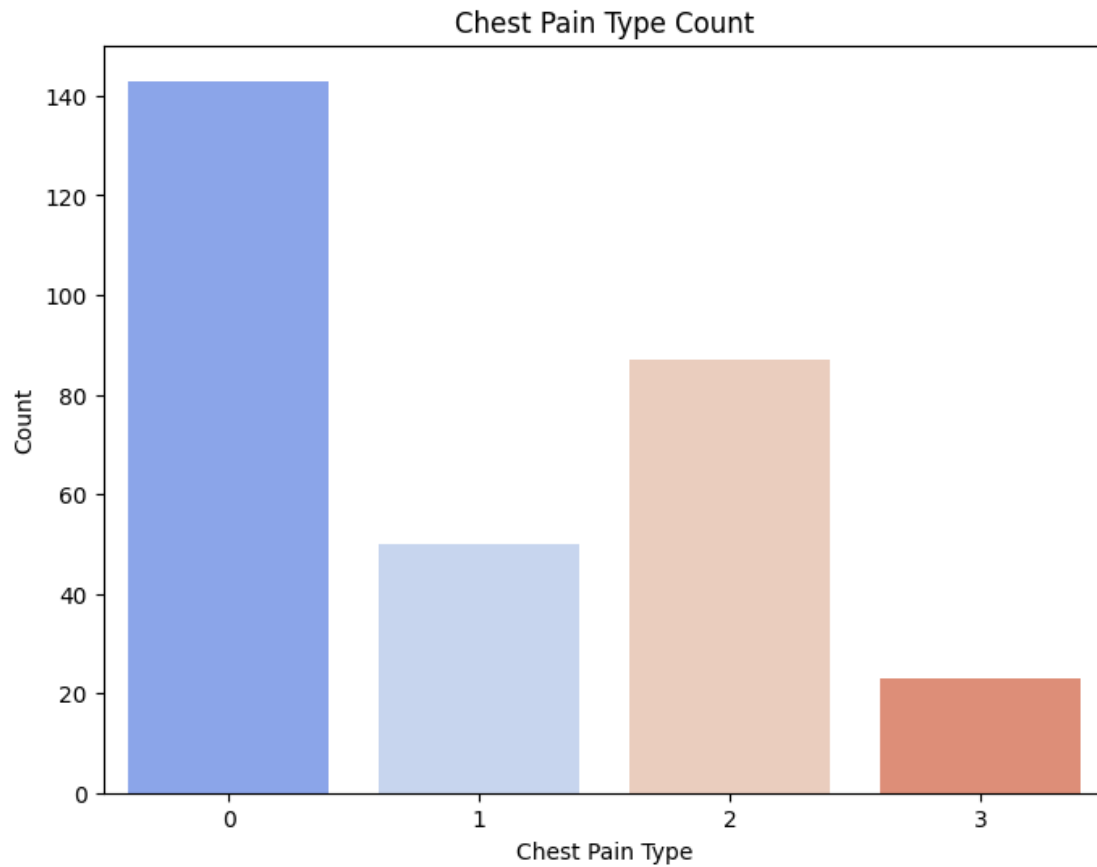


0.4 4)Count plot

```
[10]: #Chest Pain Type Count
plt.figure(figsize=(8,6))
sns.countplot(data=df, x="cp", palette="coolwarm")
plt.title("Chest Pain Type Count")
plt.xlabel("Chest Pain Type")
plt.ylabel("Count")
plt.show()
```

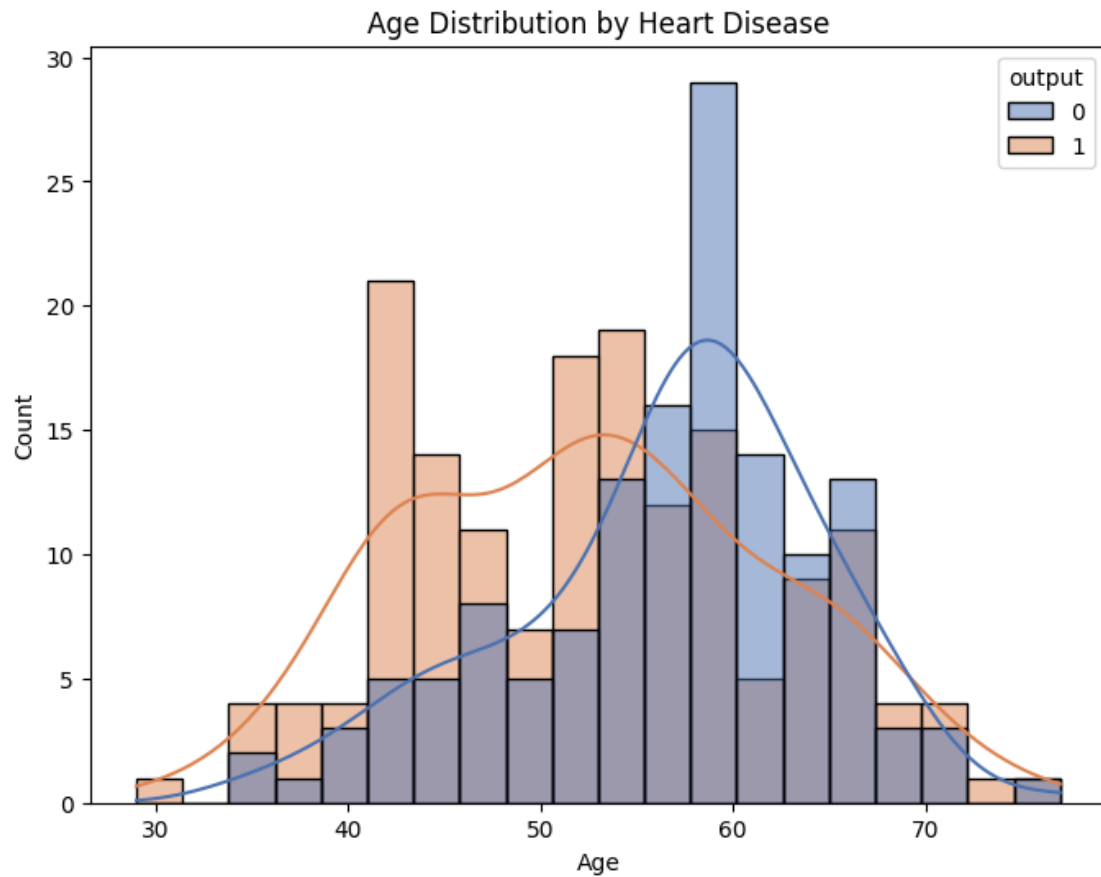
C:\Users\Yash\AppData\Local\Temp\ipykernel_54992\1648724664.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.



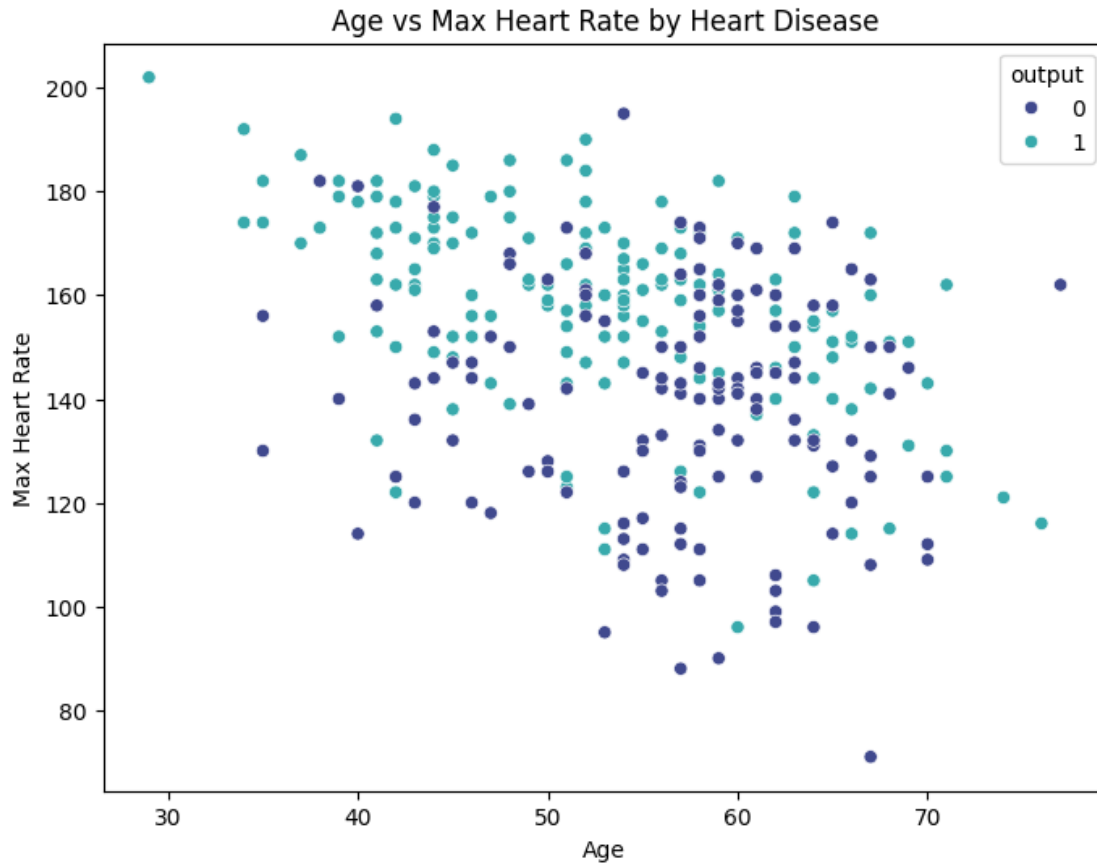
0.5 5)Histogram

```
[11]: #Age Distribution by Heart Disease (Seaborn)
plt.figure(figsize=(8,6))
sns.histplot(df, x="age", hue="output",
             kde=True, bins=20, palette="deep")
plt.title("Age Distribution by Heart Disease")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



0.6 6)Scatter plot

```
[12]: # Age vs Max Heart Rate (Seaborn)
plt.figure(figsize=(8,6))
sns.scatterplot(data=df, x="age", y="thalachh",
                hue="output", palette="mako")
plt.title("Age vs Max Heart Rate by Heart Disease")
plt.xlabel("Age")
plt.ylabel("Max Heart Rate")
plt.show()
```

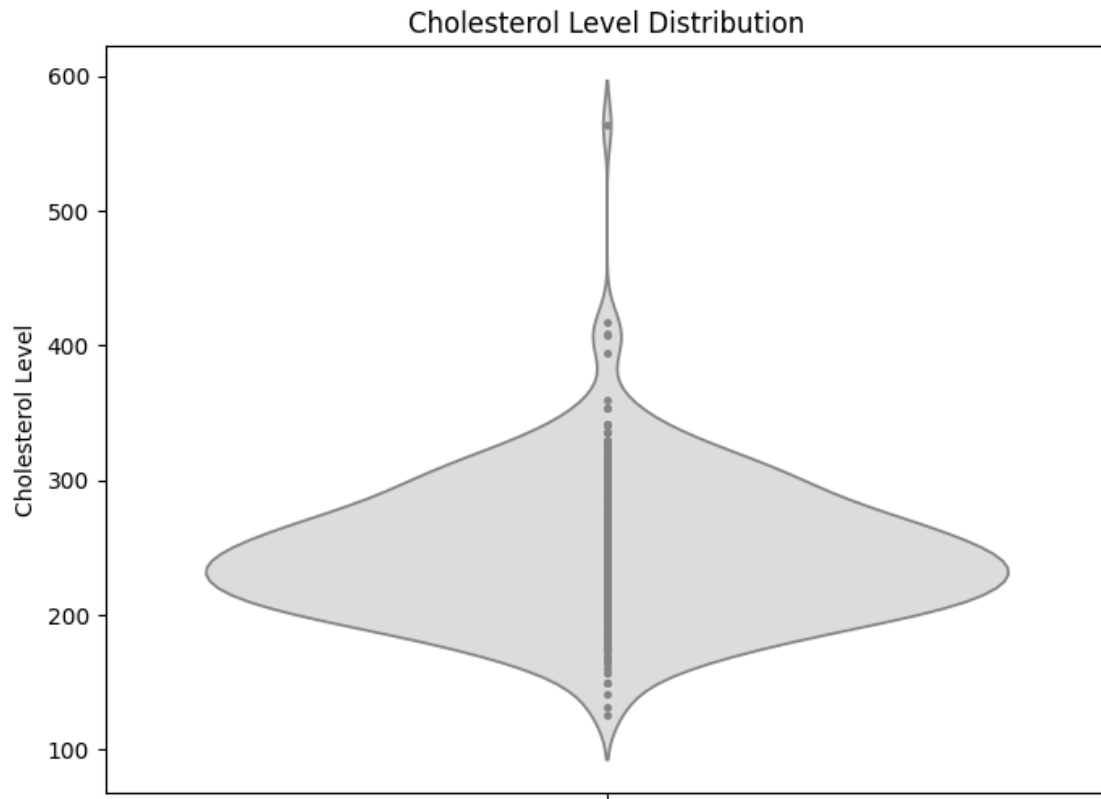



0.7 7) Violin Plot

```
[13]: # Violin Plot - Cholesterol Level Distribution (Seaborn)
plt.figure(figsize=(8,6))
sns.violinplot(data=df, y="chol",
               inner="point", palette="coolwarm")
plt.title("Cholesterol Level Distribution")
plt.ylabel("Cholesterol Level")
plt.show()
```

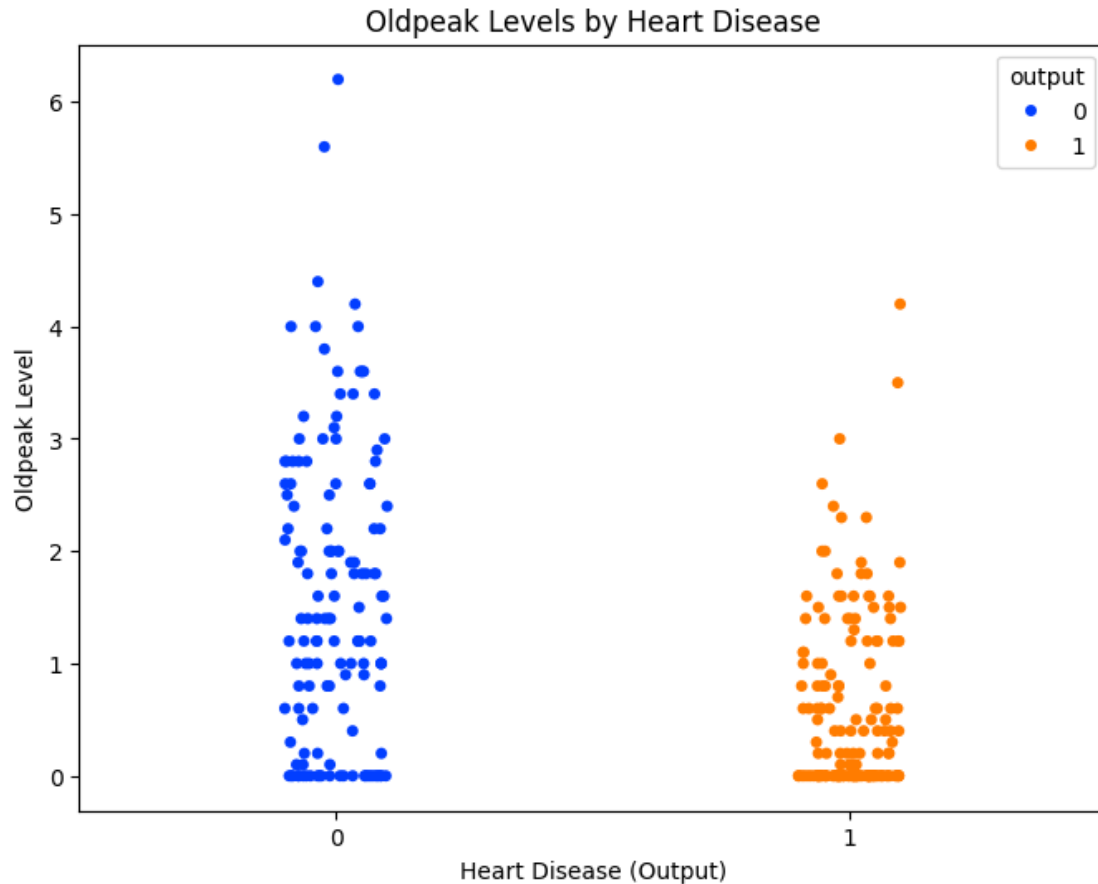
C:\Users\Yash\AppData\Local\Temp\ipykernel_54992\3171357582.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.



0.8 8)Strip plot

```
[14]: # Strip Plot - Oldpeak Levels by Heart Disease (Seaborn)
plt.figure(figsize=(8,6))
sns.stripplot(data=df, x="output", y="oldpeak",
              hue="output", palette="bright", jitter=True)
plt.title("Oldpeak Levels by Heart Disease")
plt.xlabel("Heart Disease (Output)")
plt.ylabel("Oldpeak Level")
plt.show()
```

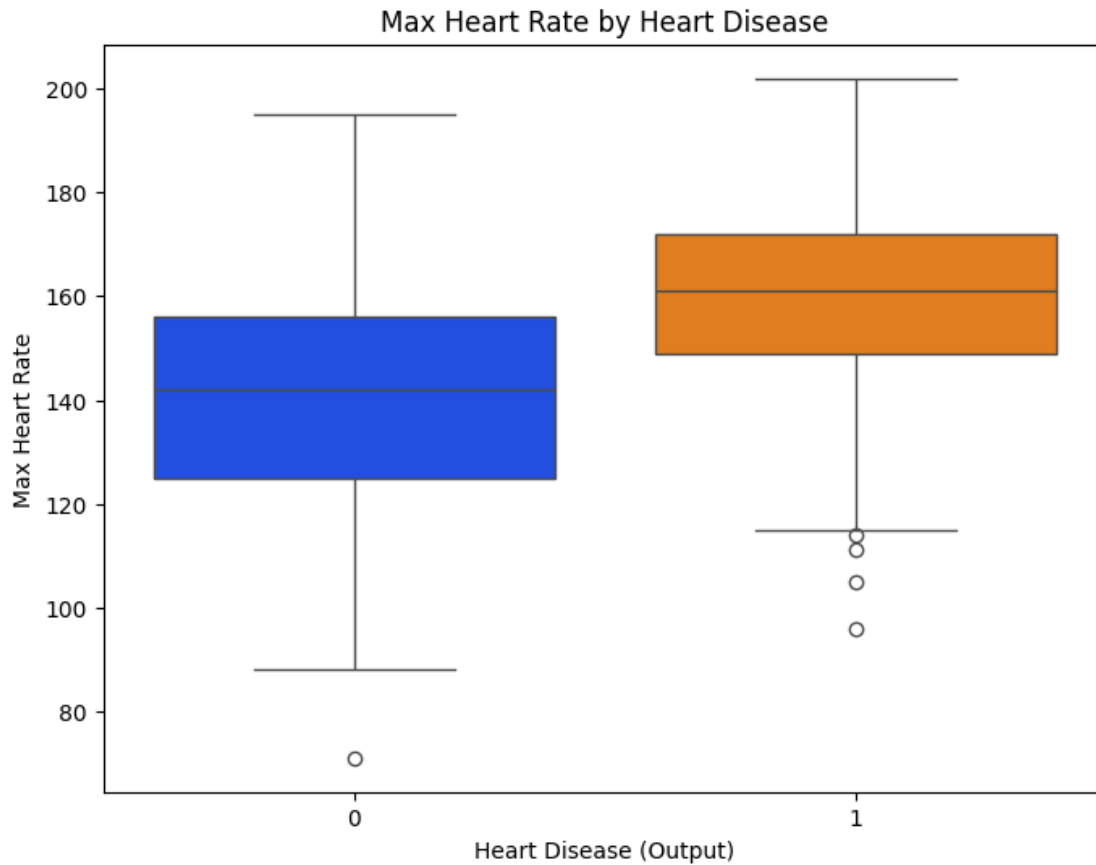


0.9 9)Box plot

```
[15]: # Box Plot - Max Heart Rate by Heart Disease (Seaborn)
plt.figure(figsize=(8,6))
sns.boxplot(data=df, x="output", y="thalachh",
            palette="bright")
plt.title("Max Heart Rate by Heart Disease")
plt.xlabel("Heart Disease (Output)")
plt.ylabel("Max Heart Rate")
plt.show()
```

C:\Users\Yash\AppData\Local\Temp\ipykernel_54992\3568270867.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.



0.10 10) Cluster plot

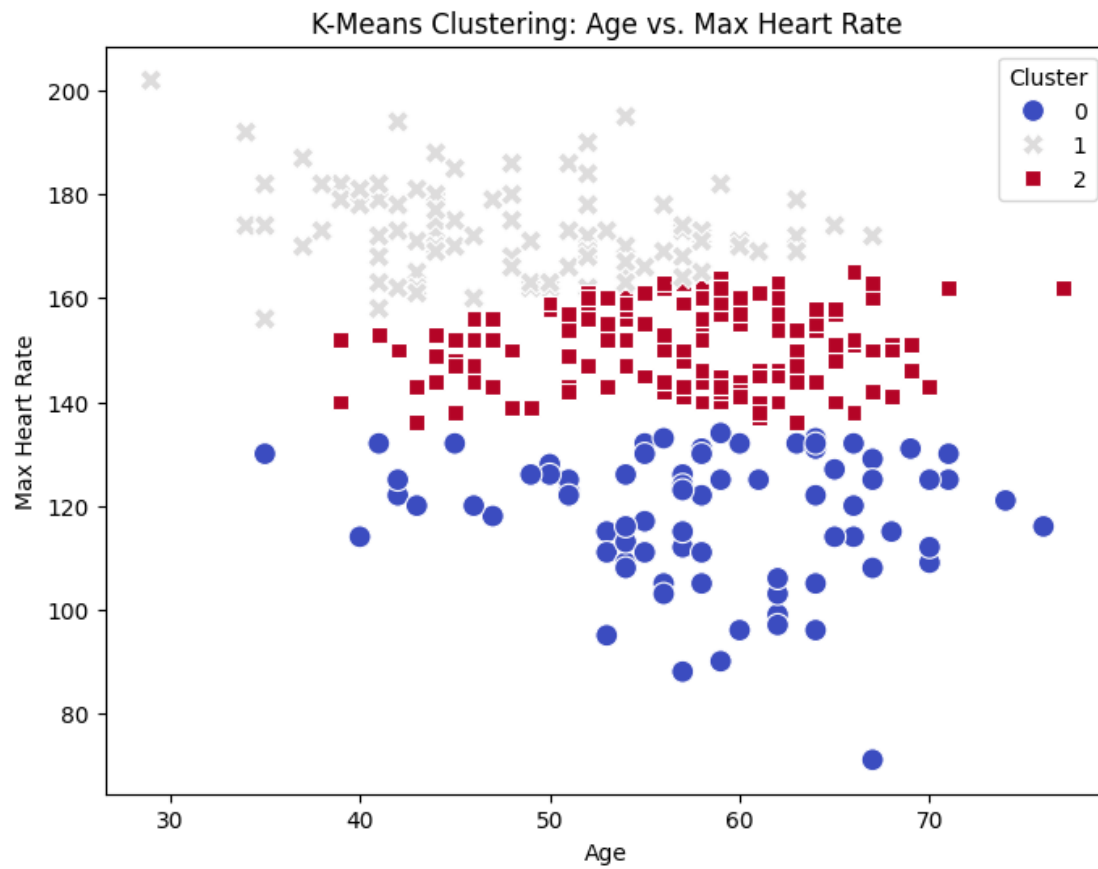
```
[16]: from sklearn.cluster import KMeans

# Selecting features for clustering
X = df[['age', 'thalachh']]

# Applying K-Means clustering
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df['cluster'] = kmeans.fit_predict(X)
```

```
[17]: # Scatter plot with clusters
plt.figure(figsize=(8,6))
sns.scatterplot(data=df, x="age", y="thalachh",
                hue="cluster", palette="coolwarm",
                style="cluster", s=100)
plt.title("K-Means Clustering: Age vs. Max Heart Rate")
```

```
plt.xlabel("Age")
plt.ylabel("Max Heart Rate")
plt.legend(title="Cluster")
plt.show()
```



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